Subsystems as Aspects

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Abstract

This paper bridges two seemingly unrelated topics: (1) the metaontological problem of talking about "portions of reality" without committing to a specific object-property analysis of these "portions", and (2) David Wallace's subsystem-recursive view of physical theories. Metaontologists debate whether mereological disputes over composite objects are merely verbal (Hirsch, 2005) or reflect genuine disagreements about how the world is (Sider, 2009). Both sides often pose this debate as being about whether there is an objectively correct way to carve "portions of reality" into objects and properties, but they treat this "portion" talk as merely metaphorical. I show how we can make it precise by interpreting portions of reality as aspects of facts, as understood in Rayo (2025). The resulting formalism is neutral with respect to any particular object-property analysis of those "portions". This same formalism also yields an interpretation of Wallace's (2022a) subsystem structures by identifying each physical subsystem in a subsystem structure with a corresponding aspect, thereby supporting the thesis that the physical treatment of subsystems can remain neutral in exactly the same way. This provides fresh support for structural realist positions defended in the philosophy of physics (Ladyman and Ross, 2007; Wallace, 2022b).

1 Introduction

Both metaontologists and philosophers of physics face a version of the same challenge. The challenge is to talk about a "portion of reality" or a "subsystem" without committing to a particular analysis, or "carving", of that portion or subsystem into objects and properties. In both fields, the challenge arises from claims that rival descriptions are equivalent, and such claims concern a *representational target*: something the rival descriptions are descriptions *of*. This raises a question: what is the minimal structure in the world that these claims require?

In metaontology, this challenge arises when deflationists claim that different sides of a metaphysical debate have equivalent descriptions of "the same portions of reality", despite appearing, on the surface, to commit to different objects and properties. Mereological nihilists[1] deny that particles arranged table-wise compose a further thing—namely, a table—(Van Inwagen, 1995), while mereological universalists affirm that they do (Lewis, 1991). *Deflationists* like Hirsch (2002; 2005; 2009) argue that such disputes are merely verbal: both sides describe the same facts in slightly different languages. *Inflationists* like Sider (2009) argue that, on the contrary, at most one side carves reality at its joints.[2] Both sides often frame the debate as being about whether there is an objectively correct way to carve "portions of reality" into objects and properties: is the "tabley portion of reality" a *table*, or is "it" merely *some particles arranged table-wise*? But this talk sits uneasily with deflationism, since it appears to quantify over portions as individuals (Sider, 2023). So far, every attempt at cashing out this notion either smuggles in a particular object-property analysis, or fails to capture the deflationist's claim that both sides describe the same portion of reality.

In philosophy of physics, the challenge takes a different but analogous form. Structural realists hold that what physics describes is *structure*—something that can be presented through different but equivalent mathematical formulations without any single formulation being privileged (Ladyman and Ross, 2007; Wallace, 2022b). But what makes two formulations "equivalent"? On a "language-first" view of theories, equivalent theories talk about the same entities and say the same things about them, just using different words. On the "math-first" view favored by Wallace (2022b), equivalence is more permissive—something like a structure-preserving transformation between models. Yet, as Wallace himself illustrates, mathematically equivalent theories routinely "mix up ontological categories": there is often no simple mapping from the objects of one formulation to those of another, and likewise with the properties and relations. So a structural realist will follow Wallace in insisting that equivalence cannot be construed as talking about the same entities and ascribing the same properties to them. But mathematical equivalence alone does not settle the question of when two formalisms describe the same physical reality. The same mathematical structure can be used to model different aspects of reality, and different—but isomorphic or dual—structures can be used to model the same aspects. What is missing is an account of what a "portion of physical reality" is that does not presuppose the object-property analysis furnished by any single formulation. Thus, the structural realist's claim that different formalisms describe the same physical situation has not yet been adequately stated.

[1] Even though I cite Van Inwagen (1995) as an example of a *nihilist*, this isn't entirely right. He is most properly understood as an *organicist*: someone who thinks particles only compose living beings. However, for the purpose of this paper, this difference won't be relevant.

[2] Sometimes "inflationism" is used to refer to a different view, according to which there are more objects than we might think, but I am not using it in this sense here. Rather, inflationism might be seen as the view that there are more *metaphysically substantive disagreements* than we might think.

I argue that these two challenges are, at bottom, *the same problem*: the problem of talking about an aspect of reality that multiple descriptions can render salient, without forcing a particular object-property analysis of said aspect. I then propose a unified resolution by interpreting both "portions of reality" and physical subsystems as *aspects of facts*, a notion developed by Rayo (2025) in the context of a theory of reference.

On this picture, the same fact can be analyzed into objects and properties in different ways. For instance, the fact that Socrates died can be analyzed into SOCRATES and HAVING DIED, or into SOCRATES'S DEATH and HAVING OCCURRED, by different modes of analysis. Each of these analyses *renders salient* different aspects of this fact by putting it against a different background of comparisons between facts. The key move is to recognize that aspects like SOCRATES are rendered salient by *comparisons between facts*: this aspect is, among other things, what the fact that Socrates died and the fact that Socrates was Greek have in common. Likewise, HAVING DIED is, among other things, what the fact that Socrates died and the fact that Plato died have in common. But nothing in this picture forces us to consider SOCRATES and HAVING DIED as *having different ontological categories*.

I take advantage of this latter point to articulate the notion of an *aspect* in a way that is independent of specific object-property ontologies, in the sense that different object-property analyses can yield the same set of aspects. Next, I formalize this notion in a higher-order logical framework. The resulting notion is neutral with respect to object-property analyses by construction: an aspect is neither an entity (type e), nor a plurality (type ee), nor a property of entities (type $e \to t$), but an operator of type $t \to t$ that filters a proposition down to what it says about a given subject matter, and we can make sense of items of different syntactic categories "corresponding" to the same aspect. I then show how this notion can be used to resolve both challenges. Accepting the resulting account does not, however, *commit* one to deflationism, either in metaontology or in philosophy of physics; it provides grounds neutral enough to *understand* the deflationist's position, and that of her opponents.

The paper goes as follows. I first present both challenges in detail in §2 and §3. I then invoke the notion of an aspect as a response to both in §4, and show how it addresses the metaontologist's challenge in §5 and the structural realist's in §6. In §7, I formalize aspects in a higher-order logical framework. In §8, I review a mathematical framework for physical theories developed in Wallace (2022a), and in §9 I show that its subsystems can be interpreted as aspects of facts. I discuss the broader implications of this correspondence in §10, and argue that the two applications reinforce one another.

2 Portions of Reality

The problem of characterizing the representational target is best understood via an example, like the debate between *mereological nihilists* and *mereological universalists*. Consider a situation in which some particles are arranged table-wise. Universalists affirm that in such a situation, there is an additional thing they compose, namely a table, while nihilists deny this. More generally, universalists claim that whenever some particles are arranged in a certain way, there is a further object that they compose, while nihilists deny this. Of course, there are positions that are intermediate between these two extremes (Markosian, 1998), but for the present purposes, it is enough to consider these two.

Some metaphysically *deflationist* philosophers such as Eli Hirsch (2002; 2005; 2009) argue that these sorts of existence debates are often *merely verbal*: each side expresses the *same* facts in a slightly different language, but there is no substantive disagreement about what the underlying facts are. For example, Hirsch contends that when the universalist describes one fact by saying "There is a table here", the nihilist might describe *the same fact* by saying "Some particles are arranged table-wise here", so their disagreement is "merely verbal".[3] Metaphysical *inflationists* like Sider (2009), by contrast, are more inclined towards views that favor one of the sides in these debates: *even if* both the nihilist and the universalist are describing the same fact, at most one of them is describing this fact in the "metaphysically correct" way, and that correctness is *determined by the world*.[4]

Often, both inflationists and deflationists feel compelled to frame their discussion precisely in these terms: as being about whether there is an objectively correct way to "carve portions of reality" into objects and properties. Indeed, Hirsch (1978) already frames discussions like this in terms of when we are "compelled to 'dignify' certain portions of reality as unitary things", while Sider (2023) pictures the structure at issue as "consisting of propertied 'portions of reality,' which quantifier meanings 'divide into entities' in different ways". To see why this talk seems to naturally capture the disagreement between the deflationist and the inflationist, consider the following questions:

(Q1) **Universalism vs. Nihilism:** Is a *tabley portion of reality* a table, or is it *merely* some particles arranged table-wise?

[3]There is a complication with describing the situation this way. What Hirsch would *really* say is that the nihilist and the universalist have inequivalent theories *when understood in English*. So Hirsch's claim isn't exactly an equivalence claim. Rather, it *depends* on one. What he would say is that there are *imaginary linguistic communities* that speak like the nihilist and the universalist—speakers of Nihilese and Universalese, respectively. Thus, it is the theories in Nihilese and Universalese that are equivalent, not the theories as understood in English. Moreover, both the universalist and the nihilist are erroneously taking themselves to speak their associated idiolect and really agreeing on the underlying facts to be described, hence the "merely verbal" character of the debate. But for the present purposes, it pays to overlook these details.

[4]There is another kind of inflationist, who flatly denies that the nihilist and the universalist are talking about the same fact even as interpreted in their own "idiolect". I don't think this position is implausible, but I won't consider it for the purpose of this discussion.

(Q2) **Deflationism vs. Inflationism:** Is there a correct object-property analysis of a *tabley portion of reality*? That is, should we assume that there is a *correct* way of answering (Q1)?

Yet this talk seems to be in tension with the spirit of deflationism: if a "portion of reality" is a *thing*, then this already favors the universalist in a sense, and if "it is not a thing", then what "is" "it"? Sider (2023) already points out that this talk "appears to involve quantification over individuals"—quantification which at least one side of the debate cannot accept, namely the nihilist—and concludes that it "can't be used in a load-bearing description of trans-ontic structure". Thus, the situation seems to favor the universalist, in some sense, and thereby also the inflationist.

The deflationist can try to make sense of this talk in a number of alternative ways, but all of these attempts seem to also bring her trouble. For instance, she may talk about regions of spacetime to capture portion-of-reality-talk (Van Inwagen, 1995; Hirsch, 1978), but that commits one to an ontology of spatiotemporal regions which themselves may be construed as fusions of spacetime points—precisely the kind of mereological baggage the nihilist was keen to avoid.[5] Alternatively, she may use *plural quantification* over particles (Boolos, 1984), recasting "portions of reality" as pluralities of particles. But this move arguably grants the nihilist's analysis an advantage: (Q2) becomes the question of whether the correct analysis of *the particles arranged table-wise* is that same plurality, or *the table*. More generally, if one tries to resolve portion-of-reality-talk by adopting an object-property analysis from the outset, that analysis will favor whichever side it happens to match (universalist, nihilist, or some other side thereof), thereby undermining the deflationist's attempt at neutrality and ultimately favoring the inflationist. Thus, it seems all cards are stacked against the deflationist in this case.

When confronted with such a situation, the deflationist might be inclined to simply give up on this talk altogether and default to some other semantic analysis of neutrality. Indeed, Hirsch (2009) eschews this talk, and articulates his claim as the claim that the nihilist and the universalist might be conceived as speaking "alternative languages"—languages that express the same set of characters[6], but distributed differently over their sentences. While this allows one to say that a universalist's "there is a table here" and a nihilist's "there are particles arranged table-wise here" describe the same fact in different languages, it fails to capture the sense in which they also single out *the same portion of reality*. After all, one might want to say that when the universalist says "There is a table here. It's round, wooden, …" and the nihilist replies "Some particles are arranged table-wise here. They form a round shape, are arranged wood-wise…", they are *not only* talking about the same facts, but talking about the same *chunk* of reality. Hence, if Hirsch is correct in claiming that universalists

[5]Moreover, some archetypal universalists—e.g. Lewis (1991)—sometimes treat their *entire* (concrete) ontology as nothing but spatiotemporal regions composed by spacetime points.

[6]A *character* is a function from contexts to sets of possible worlds.

and nihilists have equivalent pictures of the world,[7] then that equivalence runs deeper than the idea of them asserting the same facts in different vocabularies.

The problem might then be diagnosed in the following way: what is missing from the deflationist's (and the inflationist's!) picture is a way of making sense of the representational target of the debate in such a way that the debate can be understood, but without favoring either side of the first-order dispute about (Q1) from the outset—that is, without presupposing a particular object-property analysis of the representational target.

3 Structural Realism

The problem of making sense of the representational target in debates about equivalence is not unique to metaontology. Indeed, it also arises in philosophy of physics. To see why, consider *structural realism* (SR): the view that the only commitment of our scientific theories is to the *structure* of the world, rather than a particular "ontology". In its *epistemic* form (ESR), structural realism says that *we can only know* the structure of the world, and not the particular ontology that realizes such structure. In its *ontic*, and most radical, form (OSR), structural realism says that fundamentally, *structure is all there is* (Ladyman, 1998; Ladyman and Ross, 2007; Wallace, 2022b).

As Wallace (2022b) notes, there is an immediate problem with even *stating* what structural realism is supposed to say, especially in its ontic variety. Wallace's preferred formulation—the one I find most compelling among the bunch—starts by putting forth a particular conception of scientific theories. According to this conception, a scientific theory is not a collection of sentences in some language—the "language-first" view—, but rather a collection of *mathematical structures* that are used to model the target phenomena—the "math-first" view. This, in turn, induces a conception of *equivalence*; theoretical equivalence under this conception will be "something like equivalence by the standards of mathematics: a 1:1 transformation between the models that preserves mathematical structure" (p. 353). The difference between the language-first and the math-first view is then explained by a series of examples that are, supposedly, equivalent under the math-first view, but not under the language-first view. Here is a particular example which we will discuss in more detail later (Wallace, 2022b, p. 354):

> ***Non-Relativistic Quantum Mechanics***. A model of ("N-particle") non-relativistic quantum mechanics, at each time t, is given by any of the following models (among others):

[7] I must confess I am *not* exactly a deflationist in Hirsch's sense: I wouldn't use the term "merely verbal dispute" to describe the situation between nihilists and universalists. Nevertheless, Hirsch's metaontological view bears some resemblance to mine, which makes these issues worth discussing.

- An N-argument function ψ taking N *points* $\mathbf{x}_1, \ldots, \mathbf{x}_N$ in 3-dimensional (Euclidean) space and returning a complex number.
- A one-argument function Ψ taking a *single point* $\mathbf{X}$ in $3N$-dimensional space and returning a complex number.

These two structures are inequivalent when it comes to the language-first view: the first would involve variables ranging over points in 3-dimensional space, and a function of N arguments taking such variables, while the second would involve a single variable ranging over points in $3N$-dimensional space, and a function of a single argument taking such a variable. But they are inter-definable when it comes to the mathematical structure they represent. For instance, one can simply define $\mathbf{X} := (\mathbf{x}_1, \ldots, \mathbf{x}_N)$ and $\Psi(\mathbf{X}) := \psi(\mathbf{x}_1, \ldots, \mathbf{x}_N)$ to get the second structure from the first, and use certain projection functions to get the first structure from the second. Given this, Wallace considers these structures to be equivalent under the math-first view.

It is hard to deny that these structures are mathematically equivalent, in some sense. But this in itself is not a result about physics, nor about the particular discussion this equivalence claim is meant to settle. To get a hold of this, we need to understand what the equivalence claim is about. So let me introduce some context.

When it comes to the metaphysics of quantum mechanics, a question that is often discussed in the context of the Bohmian interpretation is whether the ontology of the theory should be understood in terms of a *three-dimensional* space or a *high-dimensional* configuration space. Call proponents of the former view *three-dimensionalists* (or 3D-ists) and proponents of the latter view *high-dimensionalists* (or HD-ists). Both the 3D-ist and the HD-ist think that the universe consists of particles being moved around by a wave function. The difference between them is in how they understand the *space* in which these particles move, with the 3D-ist positing *many* particles in a *three-dimensional* space (Maudlin, 2019), and the HD-ist positing *one* "marvelous particle" in a *high-dimensional* space (Albert, 2015). Naturally, the 3D-ist will pick the first structure, while the HD-ist will pick the second, as the most "straightforward" representation of their preferred ontology.

The claim that these representations are equivalent is then meant to translate into a claim about this discussion: both sides of the debate are really having a "merely verbal"—or better yet, "merely representational"—dispute. More generally, Wallace claims that in this and other cases, mathematical equivalences "mix up ontological categories", but that this does not undermine their claim to equivalence. In this sense, structural realism, in its most ambitious form, is not only about theories, but also about *the nature of ontology itself*.

But there is a problem with this view. Formal criteria alone do not settle the question of when two formalisms are equivalent—as proponents of formal approaches now concede.[8] This is independent of whether these criteria are based on a language-first or a math-first conception of theories. So the structural realist has to at least say *something* about what the extra-mathematical conditions for equivalence are. Here is where the trouble begins. An "ontology-first" philosopher has a perfectly good account for why the mathematical structures in the previous example are equivalent. Suppose the ontology of quantum mechanics *really is* that of N particles and an N-adic (physical) wave function in a three-dimensional space. Then one could describe this ontology by assigning a three-dimensional location to each term $\mathbf{p}_i$ for $i = 1, \ldots, N$, each of which tracks the location of a particle, and defining an N-argument function ψ taking N three-dimensional locations $\mathbf{x}_1, \ldots, \mathbf{x}_N$ as arguments whose value at those locations is the value of the physical wave function there. Analogously, one could stipulate that a $3N$-dimensional vector $\mathbf{P}$ will track the locations of the N particles in a certain order, and define a one-argument function Ψ taking a $3N$-dimensional vector $\mathbf{X}$ as an argument whose value at $\mathbf{X}$ is the value of the physical wave function at the points in three-dimensional space represented by certain projections[9] $\pi_1(\mathbf{X}), \ldots, \pi_N(\mathbf{X})$ to specific three-dimensional subspaces. None of this precludes there being an objective fact about *how many particles there are*, for instance, since these are merely choices of how to represent the underlying ontology. And the math-first theories are equivalent because they describe the same ontology.

So Wallace's claim does not get to the heart of the matter, since we can make the "equivalence" claim innocuous by simply saying that the mathematical structures are equivalent in virtue of describing the same ontology. Thus, for the structural realist to say what she *really* wants to say, she needs to talk about the representational target of the debate, not just the *theories* that represent this target. Moreover, this has to be done in a way that does not commit her to a particular "ontology" (in the structural realist's sense) from the outset—that is, without presupposing a particular object-property analysis of the representational target. Notice that this is *exactly* the problem faced by the metaontological deflationist.

4 Aspects of Facts

I have argued that the deflationist in metaontology and the structural realist in philosophy of physics face what is essentially the same problem: that of providing a conception of the representational target of their respective debates that does not commit them to a particular object-property analysis of said target from the outset. The problem can also be articulated by asking the following question:

[8] Dewar (2023, p. 14) grants his critics "that no purely formal criterion can be sufficient to ground judgments of equivalence, and that interpretational work is needed as well", and Halvorson (2026) lists, as the first step in "saving the program" of formal equivalence, the "Concession: No direct philosophical upshot of formal equivalence results."

[9] For the HD-ist, these projections are not obtained directly from her ontology, but from the *dynamics* of the theory.

how can we talk about an *aspect of reality* that multiple descriptions can render salient, without forcing a particular object-property analysis of it?

Ironically, a promising answer to this question comes from an analysis of object-property representation via singular terms and predicates. In Rayo (2025), Agustín Rayo develops the notion of *aspects of facts* in a discussion with Linnebo (2018) on when a singular term (like '0') truly *refers* to an object and, relatedly, on the notion of *objecthood* itself. In a highly compressed form, Linnebo argues that a term c in language L with sentential interpretation $[\![\cdot]\!]$ refers if and only if:

(i) $[\![\cdot]\!]$ respects logical entailments in a free logic,

(ii) $[\![\cdot]\!]$ assigns $\ulcorner \exists x(x = c) \urcorner$ a true proposition, and

(iii) $[\![\cdot]\!]$ interprets L such that c is governed by a predicative abstraction principle based on some partial equivalence relation $\sim$, and a suitably positioned agent with an adequate grasp of $\sim$ uses L as interpreted by $[\![\cdot]\!]$.

Rayo, by contrast, contends that only the first two conditions are required for genuine reference. To justify this position, he proposes a picture where both singular-term reference and predication are understood in terms of *rendering salient* certain aspects of facts through networks of connections. More concretely, one might analyze the fact [Socrates died] by putting it in relation to other facts, as in Figure 1. Here, the first analysis (A) picks out the object SOCRATES, while the second complementary analysis—given by the *transpose* (A^T) of (A)—picks out the property HAVING DIED. Together, they constitute a *decomposition* of the fact [Socrates died] into these two aspects.

(A)	[Socrates died] *relates to* [Socrates is wise] *and* [Socrates argues]	*as*	[Hypatia died] *relates to* [Hypatia is wise] *and* [Hypatia argues]	*and as*	[Aristophanes died] *relates to* [Aristophanes is wise] *and* [Aristophanes argues]
(A^T)	[Socrates died] *relates to* [Hypatia died] *and* [Aristophanes died]	*as*	[Socrates is wise] *relates to* [Hypatia is wise] *and* [Aristophanes is wise]	*and as*	[Socrates argues] *relates to* [Hypatia argues] *and* [Aristophanes argues]

Figure 1: A particular decomposition of the fact [Socrates died] into aspects.

But this is not the only way of decomposing [Socrates died]. For instance, one might also analyze it in terms of its relation to other facts by putting it into the following network of relations and its corresponding transpose:

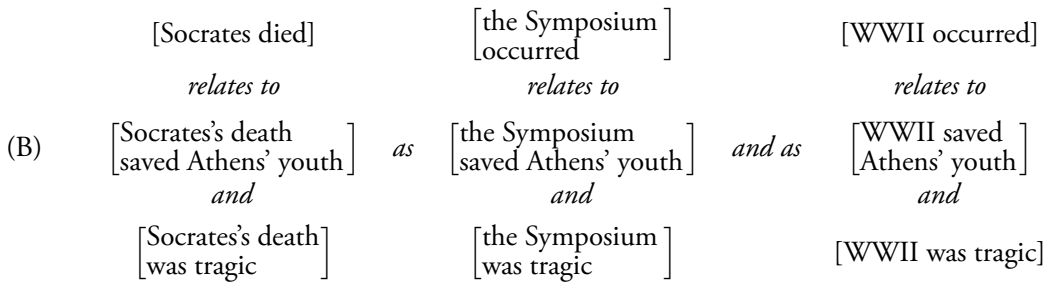


Figure 2: A different decomposition of the fact [Socrates died] into aspects.

Here, we have analyzed the same fact in terms of SOCRATES'S DEATH (via B) and HAVING OCCURRED (via the transpose B^{T}), instead of SOCRATES and HAVING DIED. Analyzing a fact under this picture can be seen as akin to decomposing a vector into its basis components: once the basis (or mode of analysis) is picked, there is an objective fact of how the vector is decomposed (or the fact analyzed), but the choice of basis (or mode of analysis) is itself conventional. Thus, *the same fact* can be analyzed using different networks of comparisons, and each analysis is *correct* relative to the network used when performing it.

5 Aspects of Facts and Portions of Reality

To see why this is helpful, consider the case of nihilism vs. universalism. Suppose a universalist fixes her attention on a particular table a, and some other objects b and c—assuming there are such things. Further, suppose the universalist uses the terms 'a', 'b', and 'c' like us. Then the universalist will be able to describe certain facts, and compare them using networks like the following (and its transpose):

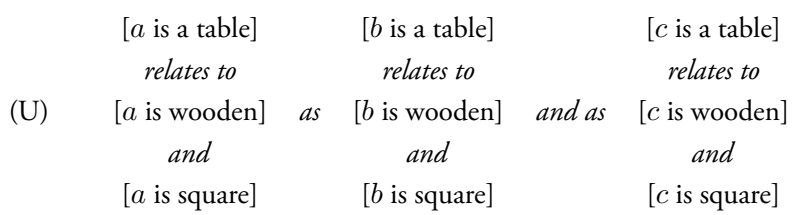


Figure 3: A network of comparisons for the universalist.

Now, the nihilist might not accept that there are such things as tables, and instead claim that there are only particles. So, in turn, she might use *plural* terms like aa that (plurally) refer to the particles that *would* compose the table (if they composed anything).[10] Similarly, she might use bb and cc for the particles that would compose b and c, respectively. Then, she might compare facts using networks like the following (and its transpose), where 'a.t.w.' is a shorthand for 'are arranged table-wise', 'a.w.w.' is a shorthand for 'are arranged wood-wise', and so on:

[10]Notice that the nihilist might also claim that it is *impossible* for aa to compose the table, so this counterfactual must not be evaluated in the standard way. Still, I hope that it makes the point clear enough.

	[aa are a.t.w.]		[bb are a.t.w.]		[cc are a.t.w.]
	relates to		*relates to*		*relates to*
(N)	[aa are a.w.w.]	*as*	[bb are a.w.w.]	*and as*	[cc are a.w.w.]
	and		*and*		*and*
	[aa are a.s.s.]		[bb are a.s.s.]		[cc are a.s.s.]

Figure 4: A network of comparisons for the nihilist.

Now, notice that, if the deflationist is correct, then 'a is a table' and 'aa are a.t.w.' are to be read as describing *the same fact* in slightly different ways, and similarly for facts in analogous positions in the networks (U) and (N). Thus, the deflationist would say that if the only facts considered are those in (U) and (N), then these are *exactly the same network of comparisons*. Moreover, one can "obtain" a from network (U) by focusing on the first column of the network, and the same goes for b and c and the corresponding columns in (U). Likewise, one can "obtain" aa from network (N) by focusing on the first column of the network, and the same goes for bb and cc and the corresponding columns in (N).

Should we then say that a and aa are *the same object*? This does not make sense, since a is *a single object*, while aa are *many objects*. What we should say instead is that 'a' and 'aa'—i.e. the *terms* used in each case—allow the nihilist and the universalist to *render salient* the same aspect of facts, namely, that corresponding to the first columns of (U) and (N). This aspect is what will ultimately correspond to the deflationist's "tabley portion of reality." Since both singular terms like 'a' and plural terms like 'aa' render aspects salient, we cannot say that an aspect is an object (i.e. the referent of a singular term), or a plurality (i.e. the referent of a plural term)—strictly speaking, neither 'a' nor 'aa' *refers* to such aspects, but rather, they are used to *render such aspects salient*.

The reader who is still skeptical of this conclusion could consider a different case. Let's focus now on the first *row* of the networks (U) and (N). The universalist might say that this row corresponds to *being a table*, while the nihilist might say that it corresponds to the plural property of *being arranged table-wise*. If the deflationist is right, then both "is a table" and "are arranged table-wise" render salient the same aspect of facts, in exactly the same way as in the previous example. Of course, *being a table* is a singular property, unlike *being arranged table-wise*, so the two expressions do not express the same condition.

Even more, both the aspect rendered salient by 'a' or 'aa' and the aspect rendered salient by "is a table" or "are arranged table-wise" are simply *aspects*, since no ontological difference has been made between them—i.e. we haven't said that one is "an object" and the other is "a property". They simply stem from comparisons between facts. So we have finally arrived at a notion that is *neutral* with respect to the object-property distinction, which was our goal all along.

6 Aspects of Facts and Structural Realism

So far, we have seen how thinking in terms of aspects helps the deflationist in metaontology make sense of "portions of reality". But, as we have seen in §3, the structural realist faces the exact same problem that we have just dealt with. So, naturally, the structural realist can benefit from the same kind of analysis.

To see how, consider the case of non-relativistic quantum mechanics treated in §3. Here, the 3D-ist posits N particles and an N-adic wave function in three-dimensional space, while the HD-ist posits a single particle and a monadic wave function in a $3N$-dimensional space. As noted before, the 3D-ist will naturally pick a mathematical model with N terms $\mathbf{p}_1, \ldots, \mathbf{p}_N$ in, say, $\mathbb{R}^3$, tracking the positions of the particles (given a certain coordinatization), and an N-argument function ψ taking N points in $\mathbb{R}^3$ and returning a complex number. Analogously, the HD-ist will pick a single term $\mathbf{P}$ in, say, $\mathbb{R}^{3N}$, tracking the position of the single "marvelous particle" (given a certain coordinatization), and a monadic function Ψ taking a point in $\mathbb{R}^{3N}$ and returning a complex number.

This is all well and good, but we still need to say what kind of *facts* are represented by the mathematical models in each case. The 3D-ist will naturally consider the following kinds of facts, at any given time t:

- $[\mathbf{p}_i = \mathbf{x}]$ will be the fact that particle i is at the position represented by $\mathbf{x}$.
- $[\psi(\mathbf{x}_1, \ldots, \mathbf{x}_N) = c]$ will be the fact that the wave function of the system at the points represented by $\mathbf{x}_1, \ldots, \mathbf{x}_N$ has a value which is represented by c.

Analogously, the HD-ist will naturally consider the following kinds of facts, at any given time t:

- $[\mathbf{P} = \mathbf{X}]$ will be the fact that the "marvelous particle" is at the position represented by $\mathbf{X}$.
- $[\Psi(\mathbf{X}) = c]$ will be the fact that the wave function of the system at the (higher-dimensional) point represented by $\mathbf{X}$ has a value which is represented by c.

Now, a deflationist who wants to put forth the equivalence claim between the three-dimensional and high-dimensional models will need to say that, in some sense, the facts represented by the two models amount to the same thing. These claims about factual identity can be construed as follows from the 3D-ist's ontology to the HD-ist's ontology:

- The conjunction $[\mathbf{p}_1 = \mathbf{x}_1 \wedge \cdots \wedge \mathbf{p}_N = \mathbf{x}_N]$ is the same fact as $[\mathbf{P} = \mathbf{X}]$, where $\mathbf{P} := (\mathbf{p}_1, \ldots, \mathbf{p}_N)$ and $\mathbf{X} := (\mathbf{x}_1, \ldots, \mathbf{x}_N)$.

- $[\psi(\mathbf{x}_1, \ldots, \mathbf{x}_N) = c]$ is the same fact as $[\Psi(\mathbf{X}) = c]$, where $\mathbf{X} := (\mathbf{x}_1, \ldots, \mathbf{x}_N)$.

Conversely, from the HD-ist's ontology to the 3D-ist's ontology, we have the following identity claims, which involve the use of particular projection functions[11] $\pi_1, \ldots, \pi_N$ from high-dimensional to three-dimensional subspaces:

- $[\pi_i \mathbf{P} = \pi_i \mathbf{X}]$ is the same fact as $[\mathbf{p}_i = \mathbf{x}]$, where $\mathbf{x} := \pi_i \mathbf{X}$.
- $[\psi(\mathbf{x}_1, \ldots, \mathbf{x}_N) = c]$ is the same fact as $[\Psi(\mathbf{X}) = c]$, where $\mathbf{x}_i := \pi_i \mathbf{X}$.

These identity claims are analogous to the metaontological deflationist's claim that "there is a table here" and "there are particles arranged table-wise here" describe the same fact. Moreover, accepting the identity claims will drive us to accept, in turn, that the three-dimensional and high-dimensional models *render the same aspects salient*. This is because the facts mentioned above produce the same networks of comparisons. In particular, when it comes to facts concerning the positions of the particles, the 3D-ist can produce the following network by varying the values of the $\mathbf{x}_i$:

$$
\begin{array}{cccccccc}
 & [\mathbf{p}_1 = \mathbf{x}_1] & & [\mathbf{p}_1 = \mathbf{x}_2] & & [\mathbf{p}_1 = \mathbf{x}_3] & \ldots \\
 & \textit{relates to} & & \textit{relates to} & & \textit{relates to} & \\
 & [\mathbf{p}_2 = \mathbf{x}_1] & \textit{as} & [\mathbf{p}_2 = \mathbf{x}_2] & \textit{and as} & [\mathbf{p}_2 = \mathbf{x}_3] & \ldots \\
 & \textit{and} & & \textit{and} & & \textit{and} & \\
\text{(3D)} & [\mathbf{p}_3 = \mathbf{x}_1] & & [\mathbf{p}_3 = \mathbf{x}_2] & & [\mathbf{p}_3 = \mathbf{x}_3] & \ldots \\
 & \textit{and} & & \textit{and} & & \textit{and} & \\
 & \vdots & & \vdots & & \vdots & \\
 & [\mathbf{p}_N = \mathbf{x}_1] & & [\mathbf{p}_N = \mathbf{x}_2] & & [\mathbf{p}_N = \mathbf{x}_3] & \ldots
\end{array}
$$

Figure 5: A network of comparisons for the 3D-ist.

and the HD-ist can produce the following network by varying the values of $3N$-dimensional variables $\mathbf{X}_i$, and using the projection functions $\pi_1, \ldots, \pi_N$ to project the $3N$-dimensional point $\mathbf{P}$ onto three-dimensional subspaces:

But notice that, under the deflationist's assumption that the factual identity claims hold, we are driven to accept that (3D) and (HD) are simply the same network of comparisons between facts, and therefore render the same aspects salient. Analogous networks of comparisons can be produced for facts concerning the wave function of the system, and how the wave function varies over time and interacts with the particles.

Notice, also, that this analysis satisfies almost everything the proponent of the math-first view insisted on: in cases like this, we can say that the 3D-ist and the HD-ist have genuinely different

[11] From the perspective of the HD-ist, these projections arise from the dynamics of the theory.

$$
\begin{array}{lccccc}
 & [\pi_1\mathbf{P} = \pi_1\mathbf{X}_1] & & [\pi_1\mathbf{P} = \pi_1\mathbf{X}_2] & & [\pi_1\mathbf{P} = \pi_1\mathbf{X}_3] \quad \dots \\
 & \textit{relates to} & & \textit{relates to} & & \textit{relates to} \\
 & [\pi_2\mathbf{P} = \pi_2\mathbf{X}_1] & \textit{as} & [\pi_2\mathbf{P} = \pi_2\mathbf{X}_2] & \textit{and as} & [\pi_2\mathbf{P} = \pi_2\mathbf{X}_3] \quad \dots \\
 & \textit{and} & & \textit{and} & & \textit{and} \\
\text{(HD)} & [\pi_3\mathbf{P} = \pi_3\mathbf{X}_1] & & [\pi_3\mathbf{P} = \pi_3\mathbf{X}_2] & & [\pi_3\mathbf{P} = \pi_3\mathbf{X}_3] \quad \dots \\
 & \textit{and} & & \textit{and} & & \textit{and} \\
 & \vdots & & \vdots & & \vdots \\
 & [\pi_N\mathbf{P} = \pi_N\mathbf{X}_1] & & [\pi_N\mathbf{P} = \pi_N\mathbf{X}_2] & & [\pi_N\mathbf{P} = \pi_N\mathbf{X}_3] \quad \dots
\end{array}
$$

Figure 6: A network of comparisons for the HD-ist.

object-property carvings of the world, while also being able to say that they render the same aspects salient. However, this happens not merely because their mathematical models are isomorphic, or bear some purely mathematical or formal relation. Rather, it is because, as interpreted, these models produce the same network of comparisons between facts.

This last point deserves emphasis, since it addresses the gap identified in §3. The structural realist's challenge was to explain what makes two mathematically equivalent formulations describe "the same physical situation" without presupposing a particular object-property analysis. The aspect framework provides an answer: two formulations describe the same physical situation when, under the relevant factual identity claims, they produce the same networks of comparisons and thereby render the same aspects salient. Some formal relation between the formulations is necessary for this—here an isomorphism, though in other cases it may be a duality, as with Einstein algebras and relativistic spacetimes (Rosenstock, Barrett, and Weatherall, 2015)—since without it the networks of comparisons would not be the same. But no such relation is sufficient. What does the explanatory work is not the bare existence of a structure-preserving map, but the fact that, when we trace what the formulations *say* about the world, they pick out the same strands of facts.

Thus, just as the deflationist about the nihilist–universalist debate can use aspects to articulate what both sides share without privileging either object-property analysis, the structural realist can use aspects to articulate what equivalent formulations share without privileging either "predicate precisification," to use Wallace's term (Wallace, 2022b). In both cases, the representational target is an aspect of facts—a portion of reality that is rendered salient by the descriptions in question, but which is not itself analyzed into objects and properties.

So far, however, the notion of an aspect has been introduced informally. In the next section, I make it precise using the resources of higher-order logic. The formalization will also allow us to define further notions that will be useful for showing that the framework of aspects extends to a general treatment of physical subsystems.

7 Formalizing Aspects

The informal picture developed in the previous sections—where aspects are "strands" common to many facts, rendered salient by networks of comparisons—has already done useful work. It has provided a way for the metaontological deflationist to talk about "portions of reality" without privileging any object-property analysis, and a way for the structural realist to say what equivalent formulations share without reducing that shared content to the ontology of either formulation. But the reader might still be wondering whether there is a way to make this notion more precise. In this section, I make it precise by identifying aspects with a certain kind of operator in a higher-order logical framework. The formalization is not merely a tidying-up exercise: it will allow us to define notions like parthood among aspects and aspect-relative modality, and it will provide the formal setting for extending the framework to a general treatment of physical subsystems. But first, I need to introduce the necessary logical machinery.

7.1 Higher-Order Logic

Higher-order logic (HOL) is a framework that extends first-order logic by introducing a *type system* for every expression in the language, by providing, for each type σ, quantifiers that bind variables of type σ,[12] and by allowing λ-abstraction. It has become increasingly popular in metaontological and modal debates (Bacon, 2018; Rayo, 2020; Dorr, Hawthorne, and Yli-Vakkuri, 2021; Bacon and Dorr, 2024). The two most basic types are e (for singular terms, such as names, definite descriptions, or functional terms) and t (for sentences or formulas). We can also include a type ee to handle *plural* terms (e.g. "the particles," "the cats"). From these basic types, we build *functional* types $\sigma \to \tau$, whose expressions combine with expressions of type σ and yield expressions of type τ. For instance, a monadic predicate like "is a dog" has type $e \to t$, since it combines with a singular term like "Fido" (type e) to form a sentence like "Fido is a dog" (type t), while a *plural* predicate like "are arranged table-wise" has type $ee \to t$. The types of other expressions are assigned in a similar way, as shown in the table below:

Expression	**Type**
Names	e
Sentences and Formulas	t
Monadic Predicates	$e \to t$
Operators ($\neg$, $\Diamond$, $\Box$, etc.)	$t \to t$
Connectives ($\wedge$, $\to$, etc.)	$t \to (t \to t)$
$\vdots$	$\vdots$

[12] Intuitively, such quantifiers can be understood as ranging over "σ-entities". As discussed below, this gloss should not be taken too literally.

In addition to assigning types, HOL makes three central additions to first-order logic. First, it provides an *identity* relation at every type: for any expressions a and b of type σ, $\ulcorner a =_\sigma b \urcorner$ is a sentence. Thus we can write $\ulcorner a =_e b \urcorner$ ("Fido is Bruno") or $\ulcorner aa =_{ee} bb \urcorner$ ("these particles are those particles"), and so forth. Second, it provides, for each type σ, *quantifiers* that bind variables of type σ: if A is an expression of type $\sigma \to t$, we may form $\ulcorner \exists_\sigma A \urcorner$ and $\ulcorner \forall_\sigma A \urcorner$. For instance, if '$F$' is a predicate of type $e \to t$, we may write '$\exists_e F$' to say "something is F", and if 'G' has type $ee \to t$, we might write '$\exists_{ee} G$' to say "some things (plurally) satisfy G". The quantifiers themselves have type $(\sigma \to t) \to t$, so even "everything" is itself a typed expression. Finally, HOL supports *λ-abstraction*, permitting the formation of functions by "abstracting away" from expressions: if A is an expression of type τ, we may form $\ulcorner (\lambda x^\sigma .\, A) \urcorner$, which is an expression of type $\sigma \to \tau$. For example, from '$\neg Fx$' (say, "*it* isn't a dog") we may form '$\lambda x^e .\, \neg Fx$' ("isn't a dog"), or more involved expressions at higher types. For brevity, we often write $\ulcorner \forall_\sigma (\lambda x^\sigma .\, A) \urcorner$ and $\ulcorner \exists_\sigma (\lambda x^\sigma .\, A) \urcorner$ as $\ulcorner \forall x^\sigma A \urcorner$ and $\ulcorner \exists x^\sigma A \urcorner$, respectively. Types are also often omitted when they can be inferred from context.

Semantically, it is often said that expressions of type e and t denote *objects* and *propositions*, respectively, whereas expressions of type $\sigma \to \tau$ denote *functions* from the interpretation of type σ to that of type τ. While this is a helpful rough guide (for example, the semantic value of "is a dog" combines with that of "Fido"—the dog itself—to yield the semantic value of "Fido is a dog", a proposition), this picture can be misleading if taken too literally—especially for philosophers who reject robust object-level ontologies of "properties" or "propositions". For instance, a nominalist higher-order logician might be happy with sentences like

$$\exists X^{e \to t} \forall y^e \, (Xy \leftrightarrow Xy) \;\wedge\; \neg \exists x^e \, \text{property}(x),$$

which involves type-$(e \to t)$ quantification—i.e. "quantifying over properties"—but denies that the predicate "property" picks out anything if read in her sense. The reader is encouraged to consult Bacon (2023, §0.5) and Rayo and Yablo (2001) for a more in-depth discussion of these subtleties when interpreting higher-order logic and higher-order quantification.

My proposal to treat aspects of facts using a certain kind of expression in a higher-order framework should be read in the same spirit. As we will see, I will say that aspects have type $t \to t$ and I will quantify over them; but in doing this, I do not mean to suggest that this can be read as first-order quantification over some particular abstract entities.[13] Rather, I am exploiting the fact that HOL allows *different kinds* of quantification. For convenience, I will use language which suggests that aspects are like "things," but one should think of this as "loose talk" (in the sense of Bacon, 2023, §0.8) for the sake of exposition.

[13] I also do not mean to *block* such an interpretation. If one wants to interpret aspects as abstract entities, one is free to do so. But the reader who chooses to do it should not hold me accountable for any of the consequences that follow *only* from this interpretation.

7.2 The Framework of Aspects: A Preliminary Overview

With this background in place, we can now return to the idea of aspects of facts. The guiding proposal is that aspects of facts are related to facts in the same way that properties (i.e. aspects of objects) relate to objects. In type-theoretic terms, if properties are of type $e \to t$ (because they map objects to propositions), then aspects of facts should be of type $t \to t$ (since facts themselves are described by sentences). This setup deviates somewhat from Rayo's original picture[14]—where aspects like SOCRATES or HAVING DIED may carry different types depending on the "dimension" of analysis—but it is precisely the price we have to pay to make sense of portion-of-reality-talk in a deflationist-friendly way.

Concretely, on this approach, both SOCRATES and HAVING DIED are treated as type-$(t \to t)$ operators that extract or "filter" a certain strand common to many facts. For instance, SOCRATES captures what is common to facts like [Socrates died], [Socrates is wise], [Socrates argues], while HAVING DIED captures what is common to facts like [Socrates died], [Hypatia died], [Aristophanes died], and so forth. Admittedly, this analysis "deletes" some information about the semantic dimension involved (e.g. object vs. property), but it facilitates our goal of letting "portions of reality" be recognized in a way that is neutral to object-property analyses.

But note that merely saying "aspects are type-$(t \to t)$ operators" does not in itself specify *which* type-$(t \to t)$ operators count as aspects. To make progress, we can note that there is a notion which already captures what these strands of facts seem to share: the notion of *subject matter* as understood by Lewis (1988) and Yablo (2014). According to this notion, a subject matter α is an operator that takes a proposition p and returns the part of p relevant to it. For instance, if $\alpha_{\mathsf{Socrates}}$ is the subject matter that concerns Socrates, then feeding it [Socrates died and Hypatia is wise] should yield [Socrates died]. This captures the sense in which $\alpha_{\mathsf{Socrates}}$ unifies facts like [Socrates died], [Socrates is wise], [Socrates argues], and so forth, in that they are all *about Socrates*—i.e. they all satisfy $\alpha_{\mathsf{Socrates}}(p) = p$. Relatedly, we can think of α_{died} as the subject matter that concerns dying, which unifies facts like [Socrates died], [Hypatia died], [Aristophanes died], and so forth.

To see how this connects to the informal picture developed in §5, recall the networks (U) and (N) (Figures 3 and 4). There, the first *column* of each network—the facts [a is a table], [a is wooden], [a is square] in (U), and the corresponding facts [aa are a.t.w.], [aa are a.w.w.], [aa are a.s.s.] in (N)—rendered salient what we informally called the "tabley portion of reality." We can now formalize this portion as an aspect $\alpha : t \to t$ that filters propositions down to what they say about that portion. Given a fact like [a is a table and b is wooden], which concerns both the first and the second columns of (U), α should return only the part concerning the first column. Thus, we

[14] Although Rayo admits that the distinction between objects and properties is not a *metaphysical* one: "The difference between objects and properties is not a difference of 'metaphysical character': no criterion based on purely metaphysical considerations could be used to distinguish between an object and a property" (Rayo, 2025).

should have:

$$\alpha([a \text{ is a table and } b \text{ is wooden}]) = [a \text{ is a table}]$$

Notice, also, that under the assumption that (U) and (N) are the same network of comparisons, this is the same as having:

$$\alpha([aa \text{ are a.t.w. and } bb \text{ are a.w.w.}]) = [aa \text{ are a.t.w.}]$$

In other words, α captures precisely the aspect corresponding to the "tabley portion of reality"—regardless of whether we render it salient by using 'a' or 'aa'.

Crucially, α is neither an entity (type e) nor a plurality (type ee), but an operator on propositions (type $t \to t$). As we saw in §5, this is exactly what neutrality requires: facts "about the tabley portion of reality" are not analyzed beyond the fact that they comprise the range of α. Unlike previous attempts, this picture does not contain an "unrelativized" domain of objects that the universalist could claim as her own. There is, perhaps, an unrelativized domain of *aspects*—and, as noted in §7.1 this is just "loose talk"—, but neither the universalist nor the nihilist recognizes this domain as her domain of objects, pluralities, or the like.

The case is exactly analogous in philosophy of physics: aspects can "correspond" to many ontologies, in the sense that they can be rendered salient by theories referring to different objects and predicating different properties of them.

7.3 Formalizing Aspects

This usage of type $t \to t$ operators as aspects suggests a set of axioms that we can phrase in the language of higher-order logic (HOL), making use of an entailment relation $\leq$.[15]

(A1) **Weakening.** $p \leq \alpha(p)$.
What p says in total entails what p says about α.

(A2) **Monotonicity.** If $p \leq q$, then $\alpha(p) \leq \alpha(q)$.
If p entails q, then the α-filtered part of p entails the α-filtered part of q.

(A3) **Idempotence.** $\alpha(\alpha(p)) = \alpha(p)$.
Applying α again does nothing if you've already filtered p once.

I do not claim that these axioms are exhaustive, but they capture the core idea that an aspect α is a filter of propositions. The formalism is meant to be general enough to apply to any pair of speakers

[15] Here, I am using the notion of entailment in Bacon and Dorr (2024). Namely, $\leq := \lambda pq.q = p \vee q$.

with different object-property analyses of the same aspects, not just universalists and nihilists, so any axioms beyond these would have to be justified on a case-by-case basis. Now, of course the nihilist and the universalist can do more than just single out *one* aspect α. Rather, they can single out a *collection* of aspects $\mathcal{A}$, corresponding to their conception of the "portions of reality" that the world contains. This is captured by the notion of a *system of aspects*:

> ***System of Aspects***. A *system of aspects* is an item $\mathcal{A}$ of type $(t \to t) \to t$ such that any α of type $t \to t$ which satisfies $\mathcal{A}$—i.e. such that $\mathcal{A}(\alpha)$—also satisfies the axioms (A1)–(A3). We say of an aspect α that it is *in* the system of aspects $\mathcal{A}$ iff $\mathcal{A}(\alpha)$.

We might imagine that the nihilist and the universalist each have their own system of aspects $\mathcal{A}_{\mathrm{N}}$ and $\mathcal{A}_{\mathrm{U}}$, respectively, corresponding to what they take to be the "portions of reality" that the world contains. To get a sense of what these aspects are, one might imagine, for instance, that the nihilist constructs $\mathcal{A}_{\mathrm{N}}$ "in steps":

1. First, single out a collection aa of particles and determine all the propositions corresponding to the claims that can be made about them. Let P_{aa} be (the set of) these propositions.

2. Next, define the aspect α_{aa} which sets $\alpha_{aa}(p)$ to the *strongest* proposition in P_{aa} that is entailed by p: the conjunction of all the P_{aa}-propositions entailed by p, which exists provided P_{aa} is closed under arbitrary conjunction (as it is on the intended reading of "all the propositions corresponding to the claims that can be made about aa"),

   $$\alpha_{aa} := \lambda p.\, \bigwedge\big(\lambda q.\, P_{aa}(q) \wedge p \leq q\big).$$

   This is the aspect that "filters out" all information that is not about the particles aa. Add α_{aa} to the system of aspects.

3. Repeat this process for all the other collections of particles that the nihilist singles out. The resulting system should be $\mathcal{A}_{\mathrm{N}}$.

The universalist might build $\mathcal{A}_{\mathrm{U}}$ in a similar way, but using singular terms instead of plural terms. The claim that the nihilist and the universalist are talking about the same portions of reality can then be understood as the claim that $\mathcal{A}_{\mathrm{N}}$ and $\mathcal{A}_{\mathrm{U}}$ are identical, or perhaps the weaker claim that they are coextensive. This doesn't require the nihilist and the universalist to have the same singular terms or plural terms in their language, or to agree on the "correct" object-property analyses of the aspects they single out.

There are, however, some subtleties. For example, the inflationist might ask why the nihilist is allowed to use plural terms in the construction of $\mathcal{A}_{\mathrm{N}}$ when the universalist might not. In some

accounts (e.g., Uzquiano 2004), if the universalist also uses plural terms then her system can express propositions that the nihilist cannot. If that is the case, $\mathcal{A}_{\mathrm{N}}$ and $\mathcal{A}_{\mathrm{U}}$ might not be identical or coextensive. The deflationist might retort that this is not a problem: both sides have set-theoretic vocabulary and might also be able to talk about sets of simples or composites, or even sets of sets thereof, which would allow the nihilist to articulate the universalist's talk by moving up one or more levels in the (transfinite) set hierarchy. But more complications follow. For instance, the inflationist might object that the universalist and the nihilist sometimes disagree on what possibilities are even available: as Sider (1993) has argued, some worlds the universalist considers possible are impossible for the nihilist. Such disagreements might also imply that $\mathcal{A}_{\mathrm{N}}$ and $\mathcal{A}_{\mathrm{U}}$ are neither identical nor coextensive.

Nonetheless, my goal here is not to defend the thesis that the nihilist and the universalist *are indeed* talking about the same portions of reality. Rather, I want to show that we can make sense of the deflationist's claim in a way that is neutral to object-property analyses. The details of how the nihilist and the universalist construct their systems of aspects are up for grabs. The claim might apply to some idealized versions of the nihilist and the universalist, or might simply not apply to this case at all, and yet we can still make sense of this claim, and of the deflationist's talk about "portions of reality". This talk might, in turn, be useful for characterizing other debates (as I try to show in §9), or might be of independent interest (as I try to show in §7.4).

7.4 Developing the Aspect Framework

So far, I have advanced the aspect framework as a way to make sense of the deflationist's talk about "portions of reality" in a way that is neutral to object-property analyses. But once we have this framework in place, we can also use it to articulate other metaphysical notions which might be of independent interest.

For instance, we can define a notion of *parthood* among aspects. We often imagine that one portion of reality sits inside another (as a table's leg is part of the table). Here aboutness should be understood *inclusively*: the subject matter of a composite includes the subject matters of its parts, so a fact about a table's leg—that it is brown, say—is thereby also a fact about the table.[16] Accordingly, if an aspect α' is a "part" of α, then whatever a proposition says about the whole α settles, in particular, whatever it says about the part α'. This is captured by the following definition, where $\leq$ is the entailment relation:

[16]Without the inclusive reading, the definition below would face counterexamples. If α_T extracted only facts about a table T "as a whole", then for $r = [T \text{ is heavy} \wedge L \text{ is brown}]$, where L is one of its legs, we would have $\alpha_T(r) = [T \text{ is heavy}]$ and $\alpha_L(r) = [L \text{ is brown}]$, and $\alpha_L \sqsubset \alpha_T$ would require the former to entail the latter. On the inclusive reading, $\alpha_T(r) = r$, and the requirement holds trivially. Inclusiveness is also how physics treats subsystems: the state of a composite system determines the states of its parts (§8).

Parthood. α' is a *part* of α ($\alpha' \sqsubset \alpha$) iff for all propositions p, $\alpha(p) \leq \alpha'(p)$.

$$\sqsubset := \lambda\alpha'\alpha.\forall p(\alpha(p) \leq \alpha'(p))$$

One can also use an aspect α to define a notion of *equivalence* for propositions and of aspect-relativized *entailment* ($\leq_\alpha$) and *strict entailment* ($<_\alpha$).[17] In turn, we might use these notions to define relativized notions of *necessity* and *possibility*:

Equivalence and Entailment. Given an aspect α and propositions p and q, we say that:

- p is *α-equivalent* to q ($p \equiv_\alpha q$) iff $\alpha(p) = \alpha(q)$. $\quad \equiv_\alpha := \lambda pq.\alpha(p) = \alpha(q)$
- p *α-entails* q ($p \leq_\alpha q$) iff $\alpha(p)$ entails $\alpha(q)$. $\quad \leq_\alpha := \lambda pq.\alpha(p) \leq \alpha(q)$
- p *strictly α-entails* q ($p <_\alpha q$) iff $\alpha(p)$ strictly entails $\alpha(q)$. $\quad <_\alpha := \lambda pq.\alpha(p) < \alpha(q)$

Relative Modality. Given an aspect α, a proposition p is *α-necessary* iff it is α-entailed by the tautology $\top$, and *α-possible* iff it is strictly α-entailed by the contradiction $\bot$.

$$\Box_\alpha := \lambda p.\top \leq_\alpha p \qquad\qquad \Diamond_\alpha := \lambda p.\bot <_\alpha p$$

One should understand $\ulcorner\Box_\alpha p\urcorner$ as saying that p doesn't impose any constraints on the aspect α. Thus, in a sense $\ulcorner\Box_\alpha p\urcorner$ is true iff p "doesn't say anything" about α—or, rather, if it only imposes *trivially satisfiable* constraints on α. Similarly, $\ulcorner\Diamond_\alpha p\urcorner$ can be read as saying that p doesn't pose *unsatisfiable* constraints on α.

One can further develop aspect-talk by quantifying over aspects to capture whether a proposition talks about a *wider* or *narrower* range of reality than another, and if it talks about it in a *finer* or *coarser* way. This requires fixing a given system of aspects $\mathcal{A}$ and reading '$\exists\alpha$' as type-$(t \to t)$ quantification over the aspects which comprise $\mathcal{A}$.

Coverage. p *covers* q ($p \leq_C q$) iff for every aspect that q says something about, p says at least as much.

$$\leq_C := \lambda pq.\forall\alpha(p \leq_\alpha q)$$

Extension. p is a *strict extension* of q ($p <_E q$) iff p covers q and p says something about at least one aspect q says nothing about.

$$<_E := \lambda pq.p \leq_C q \wedge \exists\alpha(p <_\alpha q \wedge \Box_\alpha q)$$

[17] Entailment should be understood along the lines of footnote 15. Strict entailment should be understood thus: $< := \lambda pq.p \leq q \wedge \neg(q \leq p)$.

Refinement. p is a *strict refinement* of q ($p <_R q$) iff p covers q and p says strictly more about at least one aspect q already addresses.

$$<_R := \lambda pq.p \leq_C q \wedge \exists\alpha(p <_\alpha q \wedge \neg\Box_\alpha q)$$

In each of these definitions, the crucial idea is that we look at how p and q behave under all aspects α in a system. If we interpret a given system $\mathcal{A}$ as *the portions of reality*, then these definitions give us a way to talk about how propositions can be related to one another in terms of the portions of reality they address. For instance, it is plausible that [The ball is red] is strictly extended by [The ball is red and the cube is blue], while being strictly refined by [The ball is red and made of rubber].

8 Subsystem Structures

The aspect formalism developed in the previous sections provides a framework for talking about portions of reality in a way that is neutral with respect to object-property analyses. I now turn to the question of whether this framework can be connected to an independently motivated formalism in philosophy of physics. The connection I have in mind is to David Wallace's (2022a) framework of *subsystem structures*, which was developed to argue that physical theories should be understood as describing subsystems of the universe rather than the universe as a whole.

Wallace's own motivation for this framework is distinct from the neutrality thesis discussed in §3. Rather, his aim was to argue against the interpretative assumptions that physical theories are primarily understood as (i) exactly modelling the deepest features of reality, and (ii) describing the entire universe at once. Nevertheless, the resulting formalism turns out to be well-suited for our purposes: in §9, I will show that each subsystem in Wallace's framework can be identified with a corresponding aspect, so that the subsystem structure of a physical theory gives rise to a system of aspects in the sense of §7. This correspondence provides a new way to articulate the structural realist's neutrality thesis using the tools developed in this paper.

In what follows, I provide a brief overview of the key concepts, mainly to fix notation and terminology. The reader is encouraged to consult Wallace (2022a) for a more detailed treatment with examples drawn from physical theories.

8.1 Wallace's Subsystem Structures

Wallace's framework is based on the notions of *subsystem structures*, *state structures*, and *symmetry groups*. The first of these concepts is the most basic one, and it is defined as follows:

Subsystem structure. A *subsystem structure* is a set $\mathcal{X}$ equipped with a partial order $\prec$, along with a minimal subsystem $0_{\mathcal{X}}$ and a maximal one (also denoted $\mathcal{X}$, slightly abusing notation).

Here, we should think of $\mathcal{X}$ as a set of subsystems, and of each subsystem as corresponding to a "part" of physical reality, something that will be made precise in §9. The subsystem relation $\prec$ captures the idea that one subsystem can be embedded in another. Along these lines, two subsystems X and Y are *disjoint* if any subsystem Z such that $Z \prec X$ and $Z \prec Y$ is equal to $0_{\mathcal{X}}$.

Aside from the subsystem structure itself, each subsystem is equipped with a *state space* and a set of *restriction maps* that determine how the state of a larger system is related to the state of a smaller one. This is captured by the notion of a *state structure*:

State structure. Given a subsystem structure $\mathcal{X}$, a *state structure* for $\mathcal{X}$ is specified by:

1. An assignment S of a *state space* $S(X)$ (a set of possible states) to each subsystem $X \in \mathcal{X}$, such that $S(0_{\mathcal{X}})$ is a singleton $\{*\}$: the null system has exactly one, trivial state.[18]
2. An assignment r of a *restriction map* $r_{XY} : S(Y) \to S(X)$ to each pair $\langle X, Y\rangle$ of systems with $X \prec Y$. These maps satisfy the following conditions:
   (i) $r_{XX} = \mathrm{id}_{S(X)}$, the identity map on $S(X)$.
   (ii) If $X \prec Y \prec Z$, then $r_{XZ} = r_{XY} \circ r_{YZ}$.
   (iii) If $X \prec Y$, then r_{XY} is surjective.

   These conditions ensure that if $X \prec Y$ and $y \in S(Y)$, we can unambiguously write $y|_X$ for $r_{XY}(y)$.

As a specific example, Wallace discusses the case of *N-particle theories*, in which a subsystem is a tuple $X = \langle K, I\rangle$, where K is a subset of $\{1, \ldots, N\} \subseteq \mathbb{N}$ (representing the particles to be included in the subsystem) and I is a subset of $\mathbb{R}$ (representing the period of time for which X describes the particles). Given $X_1 = \langle K_1, I_1\rangle$ and $X_2 = \langle K_2, I_2\rangle$, we set $X_1 \prec X_2$ iff $K_1 \subseteq K_2$ and $I_1 \subseteq I_2$. That is, $X_1 \prec X_2$ iff X_2 describes all the particles X_1 does and describes them for all the times X_1 does. The state space $S(X)$ of $X = \langle K, I\rangle$ is the set of smooth maps from $K \times I$ to $\mathbb{R}^3$. For each map $x : K \times I \to \mathbb{R}^3$ of $S(X)$, one should think of $x(n, t)$ as representing the location of particle n at time t. Along these lines, the *restriction maps* $r_{XY} : S(Y) \to S(X)$ for two subsystems $X = \langle K_1, I_1\rangle$ and $Y = \langle K_2, I_2\rangle$ with $X \prec Y$ should simply be the *functional* restrictions $y \in S(Y) \mapsto y|_{K_1 \times I_1} \in S(X)$ applied to these smooth maps.

[18] One might have been tempted to set $S(0_{\mathcal{X}}) = \varnothing$, on the thought that the null system has *no* states. But that convention is inconsistent with the restriction maps: since $0_{\mathcal{X}} \prec Y$ for every Y, there must be a map $r_{0_{\mathcal{X}}Y} : S(Y) \to S(0_{\mathcal{X}})$, and there is no function from a nonempty set into the empty set. A singleton state space says the right thing: there is exactly one way for nothing to be. Alternatively, one could exempt $0_{\mathcal{X}}$ from the state structure altogether; nothing below would be affected.

As a more concrete example, consider the case of $N = 3$. The *maximal* subsystem would be $\mathcal{X} = \langle \{1,2,3\}, \mathbb{R} \rangle$, and the *minimal* subsystem would be $0_{\mathcal{X}} = \langle \varnothing, \varnothing \rangle$ (whose state space, the set of maps from $\varnothing$ to $\mathbb{R}^3$, is automatically the singleton containing the empty map). A *state* of $\mathcal{X}$ would be a smooth map $x : \{1,2,3\} \times \mathbb{R} \to \mathbb{R}^3$, where, for instance, $x(1,0) \in \mathbb{R}^3$ would represent the location of particle 1 at time $t = 0$. We can also consider the functional restriction $r_{Y\mathcal{X}}$ where $Y = \langle \{1,2\}, \mathbb{R} \rangle$ is a strictly smaller subsystem of $\mathcal{X}$. Here, $r_{Y\mathcal{X}}$ would be the functional restriction of x to the subset $\{1,2\} \times \mathbb{R}$, so $x|_Y := r_{Y\mathcal{X}}(x)$ would be the function $y : \{1,2\} \times \mathbb{R} \to \mathbb{R}^3$ such that $y(n,t) := x(n,t)$ for $n \in \{1,2\}$ and $t \in \mathbb{R}$, which would represent the location of particles 1 and 2 at each time $t \in \mathbb{R}$ (Figure 7).

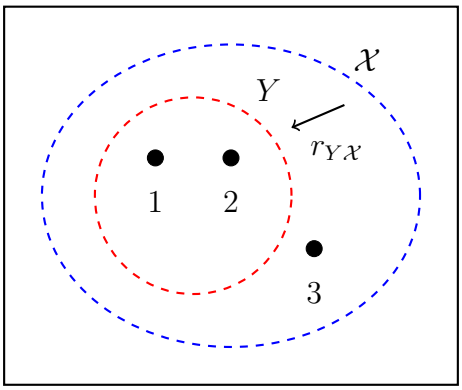


Figure 7: Subsystems $\mathcal{X}$ and Y in a 3-particle theory.

Aside from this structure, particle theories and other theories have *symmetries*: transformations between states of subsystems that leave certain physical properties invariant. Presented abstractly, symmetries can be characterized in terms of *symmetry groups*:

Symmetry group. A *symmetry group* $\mathcal{G}$ for a subsystem structure $\mathcal{X}$ (with a given state structure) consists of:

1. An assignment $\mathcal{G}$ of a group $\mathcal{G}(X)$ to each subsystem $X \in \mathcal{X}$.
2. An assignment τ of a group homomorphism $\tau_{XY} : \mathcal{G}_X(Y) \to \mathcal{G}(X)$ to each pair $\langle X, Y \rangle$ of systems with $X \prec Y$, where $\mathcal{G}_X(Y)$ is a subgroup of $\mathcal{G}(Y)$. The assignment τ satisfies the following conditions:
   (i) $\tau_{XX} = \mathrm{id}_{\mathcal{G}(X)}$, the identity map on $\mathcal{G}(X)$.
   (ii) If $X \prec Y \prec Z$, then $\tau_{XZ} = \tau_{XY} \circ \tau_{YZ}$ whenever $\tau_{XY} \circ \tau_{YZ}$ is defined.
   (iii) Each τ_{XY} is surjective.

   These conditions ensure that if $X \prec Y$ and $g \in \mathcal{G}(Y)$, then we can write $g|_X$ for $\tau_{XY}(g)$.

Each symmetry $g \in \mathcal{G}(X)$ should be thought of as *acting* on the subsystem X in a certain way, something which can be captured by the group-theoretic concept of an *action*:

Action. The *action* of a symmetry group $\mathcal{G}$ on a subsystem structure $\mathcal{X}$ is an assignment of group actions[19] $R_X : \mathcal{G}(X) \to (S(X) \to S(X))$ to each subsystem $X \in \mathcal{X}$, such that for all $X \prec Y$ and $g \in \mathcal{G}_X(Y)$, $(R_X \circ \tau_{XY})(g) \circ r_{XY} = r_{XY} \circ R_Y(g)$ or, equivalently, in more compact notation,

$$(g|_X)(y|_X) = (gy)|_X \text{ , for all } y \in S(Y).$$

In the case of N-particle theories, Wallace considers *spacetime symmetries* and *permutation symmetries*, which are respectively characterized by transformations of the following two forms:

$$(gx)(n,t) = F_t(x(n, f(t))) \text{ , and}$$
$$(gx)(n,t) = x(\sigma(n), t)$$

where $\{F_t \mid t \in \mathbb{R}\}$ is a family of bijections of $\mathbb{R}^3$, f is a bijection of $\mathbb{R}$ and σ is a permutation on $\{1, \ldots, N\}$. Here, the specific functions F_t, f depend on the particular theory being considered. Given any $X = \langle K, I \rangle$, the symmetry group $\mathcal{G}(X)$ is the set of all of these transformations $g : S(X) \to S(X)$ closed under composition, with the action being simply the application of these transformations to the smooth maps in $S(X)$.

In interpreting this formal structure, Wallace suggests states related by a symmetry should be thought of as differing only *extrinsically*—i.e. only in how they are related to other subsystems. Given this interpretative assumption, he defines the following notions of discernibility:

Discernibility. For any two states $x, x' \in S(X)$ of a subsystem X, we say that:

- x' is *intrinsically discernible* from x if there is no $g \in \mathcal{G}(X)$ with $x' = gx$, in other words, if no internal symmetry identifies them.
- x' is *extrinsically discernible* from x if it is not intrinsically discernible, and there are Y and $y \in S(Y)$, such that for any Z with $X, Y \prec Z$ and any two states $z, z' \in S(Z)$ with $z|_X = x$, $z'|_X = x'$, and $z|_Y = z'|_Y = y$, z and z' are intrinsically discernible. In this case, we say that y is a *witness* to the extrinsic discernibility of x and x'.
- x' and x are *indiscernible* if they are neither intrinsically nor extrinsically discernible.

In the next section, I propose a way to interpret these structures via the aspects framework introduced previously. This yields an interpretation of physics that remains neutral on specific object-property analyses of subsystems and physical facts about them.

[19] A *group action* of a group $(G, \cdot)$ on a set A is a function $R : G \to (A \to A)$ such that $R(g \cdot h) = R(g) \circ R(h)$ for all $g, h \in G$. We often write ga for $R(g)(a)$ when it is implicitly clear what the action is.

9 Subsystems as Aspects

Having introduced both aspects and subsystem structures, I now propose a way to bridge these two frameworks. The goal is to show how Wallace's subsystem formalism can be interpreted in terms of my aspect formalism.

Aspect interpretation. Let $\mathcal{X}$ be a subsystem structure (with its corresponding state structure). An *aspect interpretation* for $\mathcal{X}$ is a mapping such that:

1. *Aspect Assignment.* Each subsystem $X \in \mathcal{X}$ is assigned an aspect α_X, such that for all $X \prec Y$, we have $\alpha_X \sqsubset \alpha_Y$. This ensures that "smaller" subsystems correspond to "smaller" aspects in the partial order of aspects.
2. *Proposition Assignment.* Each pair $\langle X, A\rangle$, where $A \subseteq S(X)$, is assigned a proposition $p_{\langle X,A\rangle}$ such that the following conditions hold:
   (i) For all p, $\alpha_X(p) = p$ if and only if $p = p_{\langle X,A\rangle}$ for some $A \subseteq S(X)$.
   (ii) The following hold:
      (a) *Disjunction.* For all $A \subseteq S(X)$: $p_{\langle X,A\rangle} = \bigvee_{x\in A} p_{\langle X,\{x\}\rangle}$.
      (b) *Exclusivity.* For all $x, y \in S(X)$:
      if $p_{\langle X,\{x\}\rangle} \neq p_{\langle X,\{y\}\rangle}$, then $p_{\langle X,\{x\}\rangle} \wedge p_{\langle X,\{y\}\rangle} = \bot$.
      (c) *Possibility.* For all $x \in S(X)$: $\Diamond_{\alpha_X} p_{\langle X,\{x\}\rangle}$.
   (iii) For $X \prec Y$ and any $B \subseteq S(Y)$, define $B|_X = \{\, y|_X : y \in B\}$. Then we require

$$p_{\langle X,\, B|_X\rangle} \;=\; \alpha_X\big(p_{\langle Y,\, B\rangle}\big).$$

We should think of an aspect interpretation as assigning each subsystem X the aspect or portion of reality α_X that corresponds to X, with the "embedding" relation $\prec$ corresponding to the "part" relation $\sqsubset$. The proposition $p_{\langle X,A\rangle}$ is then *that X is in one of the states in A*. These propositions should comprise the range of α_X, a condition that (i) ensures. Condition (ii) spells out what it is for $p_{\langle X,A\rangle}$ to say that X is in one of the states in A: it is the disjunction of the propositions assigned to the individual states, each of which is α_X-possible, and any two of these are either identical or incompatible, since a system is in only one state at a time. The condition does not require distinct states to receive distinct propositions: mathematically distinct states may have the same physical content, which will matter for the treatment of symmetries in §9.1. Given that conjunction distributes over disjunction,[20] it follows that $p_{\langle X,A\rangle} \leq p_{\langle X,B\rangle}$ if and only if every state in A receives the same proposition as some state in B, and that $\Diamond_{\alpha_X} p_{\langle X,A\rangle}$ if and only if $A \neq \varnothing$.[21]

[20] This holds if propositions form a Boolean algebra, as is standardly assumed in higher-order metaphysics.

[21] Requiring only that $A \subseteq B$ imply $p_{\langle X,A\rangle} \leq p_{\langle X,B\rangle}$ would be too weak: it would allow one state's proposition to strictly entail another's, and the characterization of state specifications in §9.1 would fail.

Condition (iii) captures the idea that if X is a subsystem of Y, then the proposition that X is in one of the states in $B|_X$ is exactly what you get by filtering the proposition that Y is in one of the states in B through α_X. Thus, *restrictions* $r_{XY} : S(Y) \to S(X)$ in Wallace's framework correspond to applications of α_X to some proposition $\alpha_Y(p)$. Hence, the condition that r_{XX} is the identity map corresponds to the condition that $\alpha_X \circ \alpha_X = \alpha_X$, which is guaranteed by the Idempotence axiom on aspects—restricting a state to the very system it belongs to changes nothing, just as re-filtering an already-filtered proposition changes nothing. Relatedly, if we have $X \prec Y \prec Z$, then the condition that $r_{XZ} = r_{XY} \circ r_{YZ}$ corresponds to the condition that $\alpha_X \circ \alpha_Z = \alpha_X \circ \alpha_Y \circ \alpha_Z$, which obtains because $\alpha_X \sqsubset \alpha_Y \sqsubset \alpha_Z$: whenever $\alpha \sqsubset \beta$, Monotonicity applied to $\beta(p) \leq \alpha(p)$ together with Idempotence yields $\alpha(\beta(p)) \leq \alpha(\alpha(p)) = \alpha(p)$, and Weakening with Monotonicity yields the converse, so $\alpha \circ \beta = \alpha$. Lastly, the surjectivity condition corresponds to the condition that if $X \prec Y$, then for every p with $\alpha_X(p) = p$, we have a q with $\alpha_Y(q) = q$ such that $\alpha_X(q) = p$. This too obtains because $\alpha_X \sqsubset \alpha_Y$: take $q := \alpha_Y(p)$; then $\alpha_Y(q) = q$ by Idempotence, and $\alpha_X(q) = \alpha_X(\alpha_Y(p)) = \alpha_X(p) = p$ by the identity $\alpha_X \circ \alpha_Y = \alpha_X$ just established.

In sum, we can produce a mapping from each subsystem structure $\mathcal{X}$ to a *system of aspects* $\mathcal{A}_\mathcal{X}$, where each subsystem X corresponds to an aspect α_X of $\mathcal{A}_\mathcal{X}$, and each pair $\langle X, A\rangle$ with $A \subseteq S(X)$ corresponds to a proposition $p_{\langle X,A\rangle}$ in α_X's range. Of course, this mapping is not unique, and subsystem structures can be variously interpreted, but the important point is that each of these interpretations is *aspectual* in nature, and thus neutral with respect to the object-property analysis of these subsystems. This provides a new way to understand the neutrality thesis in the context of structural realism. But more than that, it also suggests that the deflationist's "portions of reality" can *themselves* be understood as subsystems, providing a new way to understand the deflationist-inflationist debate from the point of view of philosophy of physics.

9.1 Symmetries and Intrinsic Interpretations

Wallace (2022a) suggests that symmetries should be interpreted as preserving the *intrinsic* properties of a subsystem and only changing its *extrinsic* properties—i.e., its relations to other subsystems. The current framework has the resources to capture this interpretative constraint. We begin by defining the notion of a *state specification*—a proposition that exactly specifies the state of a subsystem:

> ***State specification.*** Given a subsystem $X \in \mathcal{X}$, a proposition q is a *state specification* for X iff there is a state $x \in S(X)$ such that $q = p_{\langle X,\{x\}\rangle}$.

It turns out that there is a condition that only state specifications of X satisfy, and which can be expressed without explicitly talking about states: state specifications are the *minimal* α_X-possible fixed points of α_X.

Proposition 1. *Given an aspect interpretation of $\mathcal{X}$ and a subsystem $X \in \mathcal{X}$, a proposition p is a state specification for X iff*

$$\Diamond_{\alpha_X} p \wedge (\alpha_X(p) = p) \wedge \forall q\big(\Diamond_{\alpha_X} q \wedge (\alpha_X(q) = q) \wedge (q \leq p) \rightarrow (q = p)\big).$$

Proof. Left to right: let $p = p_{\langle X,\{x\}\rangle}$. By condition (i) it is a fixed point, and by (ii) it is α_X-possible. For minimality, suppose q is an α_X-possible fixed point with $q \leq p$. By (i), $q = p_{\langle X,B\rangle}$ for some B, and $B \neq \varnothing$ since q is α_X-possible. Pick $y \in B$. By Disjunction, $p_{\langle X,\{y\}\rangle} \leq q \leq p$, so $p_{\langle X,\{y\}\rangle} = p_{\langle X,\{y\}\rangle} \wedge p$. If $p_{\langle X,\{y\}\rangle} \neq p$, Exclusivity would make this conjunction $\bot$, contradicting Possibility; so $p_{\langle X,\{y\}\rangle} = p$. Hence $p \leq q$, and $q = p$. Right to left: suppose p satisfies the displayed condition. By (i), $p = p_{\langle X,A\rangle}$ for some A, and $A \neq \varnothing$ since p is α_X-possible; pick $x \in A$. Then $p_{\langle X,\{x\}\rangle} \leq p$ by Disjunction, and $p_{\langle X,\{x\}\rangle}$ is an α_X-possible fixed point, so by minimality $p = p_{\langle X,\{x\}\rangle}$: a state specification. □

Just as we identify states via symmetries in Wallace's framework, we can "identify" state specifications via these same symmetries. This is captured by the following definition:

> ***Symmetry mapping***. Given a subsystem $X \in \mathcal{X}$, a symmetry $g \in \mathcal{G}(X)$, and two state specifications q and q' for X, we say that q *maps to* q' *under* g (written $q \mapsto_g q'$) iff there is a state $x \in S(X)$ such that $q = p_{\langle X,\{x\}\rangle}$ and $q' = p_{\langle X,\{gx\}\rangle}$.

Note that this defines a relation $\mapsto_g$ of type $t \rightarrow (t \rightarrow t)$ between propositions. We can then use this relation to define the notion of *symmetry equivalence*:

> ***Symmetry equivalence***. Given a subsystem $X \in \mathcal{X}$, we say that two state specifications p and p' for X are *symmetry equivalent* (written $p \sim_{\mathcal{G}(X)} p'$) iff there is a symmetry $g \in \mathcal{G}(X)$ such that $p \mapsto_g p'$.

Given that groups have inverses and identities, $\sim_{\mathcal{G}(X)}$ is an equivalence relation. We then use this relation to define, for any subsystem X, an aspect α_X° which is concerned only with how X is when one ignores any differences between symmetry-related states. Assuming Wallace's interpretative stance, this should correspond to ignoring any *non-intrinsic* differences between states of X. This is captured by the following definition:

> ***Intrinsic aspect***. Given a subsystem X, the *intrinsic aspect* α_X° of X is the unique subaspect of α_X[22] such that

[22] Condition (ii) makes α_X° the $\sqsubset$-greatest subaspect of α_X satisfying (i), so it is unique if it exists, by the antisymmetry of $\sqsubset$. Existence is not automatic. It holds, for instance, when α_X is generated by a map sending each world to the state of X in it, in which case α_X° is generated by sending each world to the $\mathcal{G}(X)$-orbit of that state.

(i) for all state specifications p, p' for X with $p \sim_{\mathcal{G}(X)} p'$, we have $\alpha^{\circ}_X(p) = \alpha^{\circ}_X(p')$; and

(ii) for any $\alpha \sqsubset \alpha_X$ satisfying (i), we have $\alpha \sqsubset \alpha^{\circ}_X$.

Now, we have the resources to define the notion of *discernibility* in terms of aspects.

Aspect discernibility. Given a subsystem $X \in \mathcal{X}$, we say that two state specifications p and p' for X are:

- *intrinsically discernible* ($p \; \Delta^i_X \; p'$) iff $\alpha^{\circ}_X(p) \neq \alpha^{\circ}_X(p')$;
- *extrinsically discernible* ($p \; \Delta^e_X \; p'$) iff $\neg(p \; \Delta^i_X \; p')$ and there is a subsystem Y and a state specification p_Y for Y such that for any Z with $X, Y \prec Z$ and any state specifications p_Z and p'_Z for Z with $\alpha_X(p_Z) = p$, $\alpha_X(p'_Z) = p'$, and $\alpha_Y(p_Z) = \alpha_Y(p'_Z) = p_Y$, we have $p_Z \; \Delta^i_Z \; p'_Z$.
- *indiscernible* iff neither $p \; \Delta^i_X \; p'$ nor $p \; \Delta^e_X \; p'$.

Given Wallace's interpretative constraint, we should consider indiscernible states as being *mere gauge*—that is, as states that "do not differ in any physically-relevant way and yet are mathematically represented as distinct" (Wallace, 2022a, p. 245). Since the propositions assigned to these states contain only the *physical content* of the theories that describe the subsystem structure, adopting this interpretative stance amounts to saying that indiscernible state specifications are *identical*. That is, it amounts to accepting the following constraint:

Gauge. For all $X \in \mathcal{X}$ and all state specifications p, p' for X, we have

$$(p \; \Delta^i_X \; p') \vee (p \; \Delta^e_X \; p') \vee (p = p').$$

Since condition (ii) does not require distinct states to receive distinct propositions, Gauge is consistent with it: indiscernible states keep their distinct mathematical representations, while the propositions assigned to them—their physical content—coincide, in line with Wallace's description of such states as physically the same but mathematically represented as distinct. The characterization of state specifications in Proposition 1 is unaffected, except that a state specification now pins down a state only up to sameness of content. Gauge also propagates downward: if $X \prec Y$ and $p_{\langle Y,\{y\}\rangle} = p_{\langle Y,\{y'\}\rangle}$, then condition (iii) gives $p_{\langle X,\{y|_X\}\rangle} = p_{\langle X,\{y'|_X\}\rangle}$, so identifying two states of a system identifies their restrictions to each of its subsystems.

In sum, the aspect interpretation of subsystem structures provides a way for proponents of the "ontological neutrality" of physics to articulate their thesis: physical theories are neutral with respect to object-property analyses of the subsystems they describe, because each subsystem can be identified

with an aspect of facts, and aspects are neutral by construction. Moreover, the correspondence runs in both directions. Not only can subsystems be seen as aspects, but the deflationist's "portions of reality" can be seen as subsystems—entities with state spaces, restriction maps, and symmetry groups—providing concrete, physically grounded instances of the aspect framework.

This correspondence also suggests a natural strategy for revisiting debates like the one between the 3D-ist and the HD-ist discussed in §6. There, I argued informally that, conditional on the relevant factual identity claims, the two formulations produce the same networks of comparisons and thereby render the same aspects salient. The subsystem structure formalism now offers a setting in which this argument could be made precise: one would interpret each formulation as a subsystem structure, apply the aspect interpretation to both, and verify that the resulting systems of aspects are coextensive. I leave the details of such a construction for future work, but the framework developed in this paper shows what the equivalence claim would amount to—not bare mathematical isomorphism, but identity of the aspects rendered salient—and provides the formal tools with which to establish it.

10 Conclusion

This paper began with a question: what is the minimal structure in the world that equivalence claims in metaontology and philosophy of physics require? The answer I have developed is that the relevant representational targets are *aspects of facts*—strands of reality that can be rendered salient by different descriptions without being reducible to the object-property analysis furnished by any single description.

The framework was motivated by two independently arising challenges. In metaontology, the deflationist needed a way to talk about "portions of reality" without presupposing a particular carving into objects and properties. In philosophy of physics, the structural realist needed a way to say what equivalent formulations share without presupposing the ontology of either formulation. I argued that these are instances of the same problem, and that the aspect framework resolves both.

The two applications reinforce one another. The metaontological case shows that the framework is philosophically well-motivated: it arises from a genuine need in debates about the status of ontological disputes, and it achieves neutrality in a way that existing proposals—spacetime regions, plural quantification, and other surrogates for portion-talk—do not. The physical case shows that the framework is not *ad hoc*: it captures structure already implicit in a formalism developed for independent reasons in philosophy of physics. What Wallace's subsystem structures provide is a mathematical setting in which aspects can be identified with physical subsystems, so that the neutrality thesis is not merely a philosophical aspiration but a feature of the formalism itself.

What distinguishes this approach from more familiar defenses of the neutrality of physics (Ladyman and Ross, 2007; Wallace, 2022b) is that it originates from tools already used in metaontological debates about objects and properties. In that sense, it "comes from within" metaphysics, and shows how philosophers can adopt structuralist stances without discarding the conceptual tools of traditional metaphysics. Conversely, the connection to subsystem structures shows how insights from philosophy of physics can inform metaontological debates: the deflationist's "portions of reality" can be understood as physical subsystems, providing concrete instances of the aspect framework in action.

Several questions remain open. First, the axioms I have proposed for aspects (Weakening, Monotonicity, Idempotence) are deliberately minimal, and further axioms might be needed for specific applications. Second, the relationship between the aspect framework and existing formal accounts of theoretical equivalence—such as categorical equivalence or definitional equivalence—deserves further investigation. Third, as noted above, applying the subsystem-aspect correspondence to specific cases in philosophy of physics—such as the debate between three-dimensionalists and high-dimensionalists in quantum mechanics—would require constructing explicit aspect interpretations and verifying their coextensiveness. I leave these as directions for future work.

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